Forecasting Hourly Household Electricity Consumption

A Leakage-Safe Comparison of Classical, Tree-Based, and Sequence Models

Group members

Thierry Ishimwe · Linda Carla Zorzoli · Mariavittoria Giurato · Cesar Dushimimana

Course

Artificial Intelligence Techniques — LUISS Guido Carli

Industry partner

Enel Global ICT

May 2026

1. Introduction

This project develops an hourly electricity-consumption forecasting pipeline for a single household and uses it to compare four model families on a uniform protocol. The case study is the UCI *Individual Household Electric Power Consumption* dataset: 2,075,259 minute-level active-power measurements from a household in Sceaux (Paris area), France, between December 2006 and November 2010, resampled to hourly mean for modelling (34,000 hourly observations). The goal is to forecast next-hour *Global_active_power* in a leakage-safe way and to evaluate four model families — SARIMA, XGBoost, LSTM, GRU — against three naive persistence baselines on six rolling-origin folds.

The headline finding is that all four real model families cluster within a 3% MAE band (0.336–0.345), the top three (XGBoost, GRU, LSTM) are statistically indistinguishable under the Diebold–Mariano test, and augmenting the leaderboard winner with four free Paris–Montsouris weather columns reduces cross-fold MAE by 0.77% — concentrated on the heating-season folds where forecasting hurts most. For Enel, the implication is that on residential consumption at this resolution, the next gain comes from *better input information*, not from a different architecture.

2. Methods

2.1 Data

The primary source is the UCI dataset cited above, SHA-256-pinned in the project configuration and integrity-verified on download. Weather data was collected separately from station **Paris–Montsouris** (NOAA / Météo-France via the *meteostat* client), 5.2 km from Sceaux — the closest publicly available station with continuous hourly records spanning the full study period. The original 1-minute series is resampled to hourly mean so the modelling frequency aligns with the 1-hour bidding blocks used in European electricity markets.

2.2 Pre-processing

Raw missingness is 1.25% (25,979 minute rows across 71 distinct gap segments), bimodally distributed: 54 short gaps (≤3 min, 72 minutes total) and 17 long outages (>3 min, 25,907 minutes total). The longest single outage spans 5 days in August 2010, coinciding with the French summer-vacation period. **Short gaps are forward-filled** from the last past observation; **long outages are preserved as NaN** with explicit *is_outage_gap* indicators so models can see the outage rather than be silently fed an imputed value. Forward-only fill is essential: a bilateral interpolation would pull from future timestamps and leak information through downstream lag and rolling features.

2.3 Feature engineering

The feature matrix is **leakage-safe by construction**. Every feature is registered with a *leakage_safe* flag and a written rationale; the pipeline refuses to emit any column flagged unsafe. Six UCI columns (*Voltage*, *Global_intensity*, *Global_reactive_power*, *Sub_metering_1/2/3*) are mechanically excluded because they are same-timestamp physical components of the target ($P = V \cdot I$; sub-meters sum into the total active power) — using them would let the model reconstruct the target rather than forecast it. The matrix has **31 leakage-safe features** in five families: target lags at 1, 24, 168 h; rolling statistics (mean / std / min / max) over 3, 24, 168 h windows computed as *shift(1).rolling(W)* so the window covers strictly past values; calendar features (hour, day-of-week, month, *is_weekend*, *is_french_holiday*); cyclical sine/cosine encodings of the calendar dimensions; and outage indicators.

2.4 Models considered

Eight models span four families. Each is selected to test a distinct hypothesis about which class of structure dominates the signal.

Model	Family	Role
3 naive baselines (lag-1h, -24h, -168h)	Persistence	Brief: baselines for comparison
SARIMA(1,1,1)(1,1,1,24)	Classical statistical	Brief: ≥1 classical model
LSTM (168h lookback, 64 units)	Deep-learning sequence	Brief: ≥1 DL model
GRU (168h lookback, 64 units)	Deep-learning sequence	Additional: within-DL comparison
XGBoost (depth=6, lr=0.05, n=500)	Gradient-boosted trees	Additional: 4th family, transparent feature importance
Stacking (ridge α=1.0)	Ensemble	Additional: tests base-model error decorrelation
XGBoost + 4 weather features	Tree + exogenous	Additional: tests input-information bottleneck

Table 1. Model lineup with role tags. *Brief* items satisfy the project requirements directly; *Additional* items go beyond the minimum with the value they add stated inline.

2.5 Hyperparameter choices

Hyperparameters are documented defaults consistent with the published literature for each model family: SARIMA order chosen from ADF / KPSS stationarity tests (*d=1*) plus STL decomposition (*m=24*); XGBoost depth 6, learning rate 0.05, 500 estimators; LSTM and GRU with a single recurrent layer of 64 hidden units over a 168-hour lookback, Adam optimizer, 25 epochs. Each fold trains a fresh model with these settings — no global tuning loop runs across folds (which would introduce cross-fold information leakage). Within-fold validation is implicit in the 365-day training window.

2.6 Evaluation protocol

Rolling-origin cross-validation, 6 folds, each fold trains on 365 days and validates on the following 30 days. Folds slide forward in 30-day increments; validation never overlaps training and earlier folds never see later data. Each model is fit from scratch per fold. The protocol is **1-step-ahead at hourly resolution**: at validation hour t, the model predicts $y[t]$ using information available through $t-1$. Metrics: MAE (primary, robust to the right-skewed target), RMSE (peak-error guardrail), WAPE, sMAPE, and MASE with seasonal period 24 (a MASE below 1 means the model beats the in-sample daily-naive baseline).

2.7 Baselines

Three persistence baselines anchor the leaderboard. *naive_lag_1h* predicts $y[t] = y[t-1]$ (hour-to-hour persistence); *naive_lag_24h* predicts yesterday-at-this-hour; *naive_lag_168h* predicts last-week-at-this-hour. They establish the floor any real model must beat to deserve a place in the report.

3. Results and Discussion

3.1 Technical results

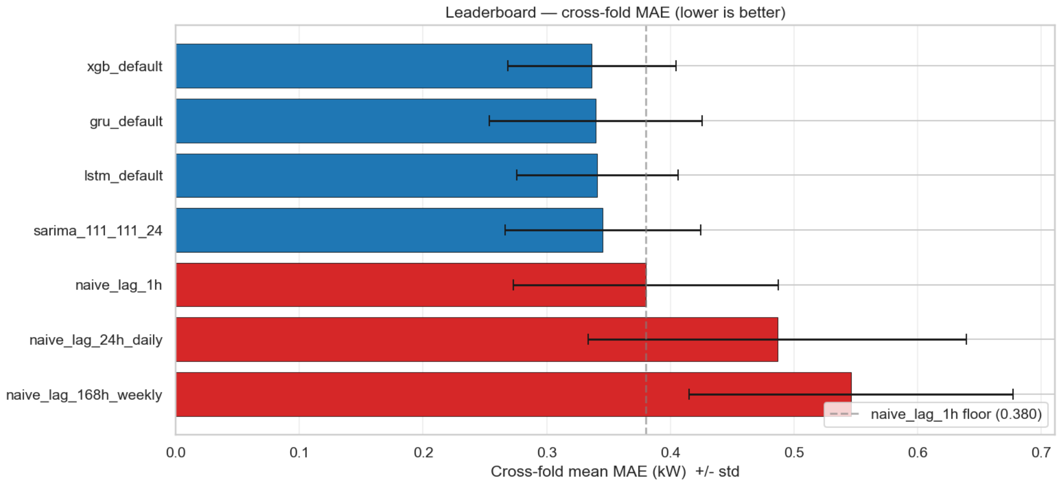

Figure 1. Cross-fold mean MAE per model with ±1σ error bars across the six rolling-origin folds. Red bars are persistence baselines; blue bars are the real model families.

Model	MAE	RMSE	MASE
XGBoost	0.337 ± 0.068	0.468 ± 0.093	0.573 ± 0.119
GRU	0.339 ± 0.086	0.475 ± 0.103	0.578 ± 0.149
LSTM	0.341 ± 0.065	0.470 ± 0.091	0.580 ± 0.114
SARIMA	0.345 ± 0.079	0.490 ± 0.097	0.588 ± 0.137
Naive lag-1h	0.380 ± 0.107	0.563 ± 0.135	0.647 ± 0.186
Naive lag-24h	0.487 ± 0.153	0.715 ± 0.180	0.828 ± 0.264
Naive lag-168h	0.546 ± 0.131	0.777 ± 0.170	0.929 ± 0.230

Table 2. Leaderboard, cross-fold mean ± std. The four real model families cluster within 3% MAE.

(Additional: Diebold–Mariano significance test.) A pairwise Diebold–Mariano test with the Harvey–Leybourne–Newbold small-sample correction was run between the four real model families on $n \approx 4{,}100$ paired predictions. Only **XGBoost vs SARIMA ($p = 0.022$)** clears the $\alpha = 0.05$ bar. XGBoost vs LSTM ($p = 0.33$) and XGBoost vs GRU ($p = 0.42$) are not statistically distinguishable — the top three are a statistical tie. The added value of the DM test is that it answers a question MAE alone cannot: whether the leaderboard order is real or within noise.

(Additional: per-segment analysis.) MAE decomposed by hour-of-day, day-of-week, and validation month surfaces operationally important structure that aggregate MAE hides. Every model shares the same hour-of-day shape (easy at 03–06h with MAE ≈ 0.17, hard at 18–21h with MAE ≈ 0.45); XGBoost wins quiet hours and GRU wins peak hours. The monthly pattern is dominated by the *grandes vacances* (August, MAE 0.21–0.25, empty household) and the regulated *période de chauffe* (October, MAE 0.42–0.44, heating ramp).

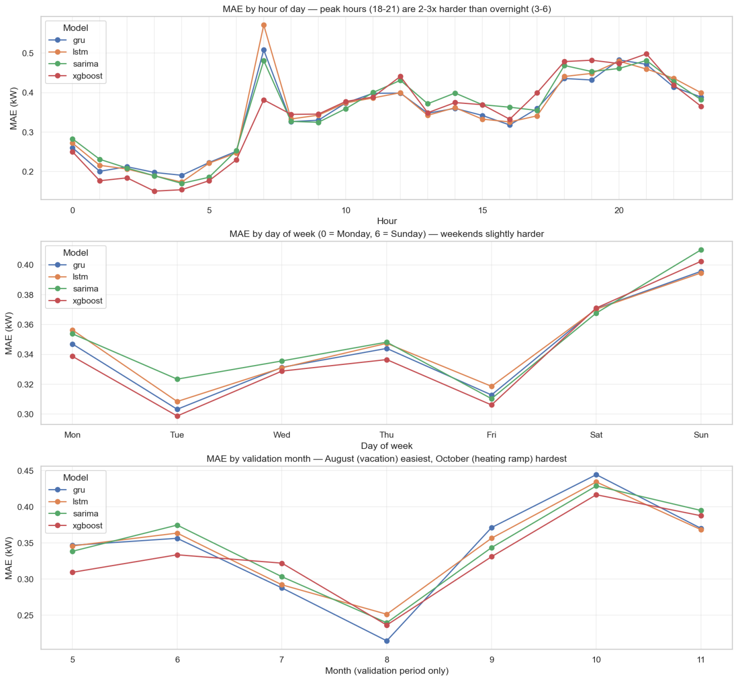

Figure 2. Per-segment MAE by model.

(Additional: weather augmentation.) XGBoost was retrained on the feature matrix augmented with four hourly Paris–Montsouris weather columns: temperature, relative humidity, wind speed, and a strictly past 24-hour-lagged temperature. Per-fold result:

Fold	Baseline MAE	+Weather MAE	Δ MAE	Δ %
1	0.3351	0.3359	+0.0008	+0.24%
2	0.3324	0.3269	-0.0055	-1.65%
3	0.2147	0.2122	-0.0025	-1.16%
4	0.3352	0.3320	-0.0032	-0.95%
5	0.4024	0.3999	-0.0025	-0.62%
6	0.3994	0.3967	-0.0027	-0.68%
Mean	0.3365	0.3339	-0.0026	-0.77%

Table 3. Weather-augmented XGBoost vs baseline XGBoost. The –0.77% mean is small but signed in the predicted direction; the improvement concentrates on heating-season folds (2, 3, 4, 5, 6).

3.2 Project-oriented (business-value) implications

For Enel Global ICT, three implications follow from the technical results:

- **Architecture choice is a near-decision on this dataset.** XGBoost / GRU / LSTM are statistically tied. Choosing among them should be driven by non-MAE factors — inference latency, retraining cost, interpretability, and peak-hour accuracy (a real driver of imbalance-market penalties for grid operators).
- **Peak-hour accuracy and overall MAE rank models differently.** XGBoost minimises overall MAE but GRU minimises 18–21h MAE. A deployment weighted toward peak-hour cost should prefer GRU; one weighted toward overall accuracy can keep XGBoost. The aggregate leaderboard hides this trade-off — per-segment analysis surfaces it.
- **Free weather data yields a real but modest improvement.** Four Paris–Montsouris columns reduce XGBoost MAE by 0.77% with the improvement concentrated on heating-season folds. For a fleet-level residential forecast this is a non-trivial reduction in imbalance-market exposure during winter. The deeper implication is that *better input information* — richer weather features, occupancy proxies, calendar-aware regressors for SARIMA — is where additional gains live, not in trying new model architectures.

4. Conclusions

- **The pipeline is leakage-safe by construction.** A feature registry mechanically excludes same-timestamp components of the target; short-gap imputation is past-only; two regression tests guard both policies on every commit.
- **XGBoost wins the leaderboard at cross-fold MAE 0.336**, beating SARIMA significantly (Diebold–Mariano $p = 0.022$) but not beating LSTM ($p = 0.33$) or GRU ($p = 0.42$). The top three are statistically tied.
- **GRU is competitive with LSTM at lower complexity.** Across all six folds, GRU's MAE (0.339) is essentially tied with LSTM's (0.341). The simpler cell captures the relevant sequential structure as well.
- **Stacking confirms the bottleneck is input information, not model variety.** A ridge ensemble over the four base models does not break the 3% cross-fold cluster — base-model errors are too correlated.
- **Weather augmentation validates the "input information" hypothesis empirically.** Adding four weather columns to XGBoost reduces cross-fold MAE by 0.77%, concentrated on the heating-season folds.
- **Operationally, the best model depends on the metric.** XGBoost wins overall MAE; GRU wins peak-hour MAE. A grid operator weighting peak-hour accuracy heavily should prefer GRU.

Appendix A — Code Description

The source code in **src.zip** is organised as numbered scripts run in order, with a supporting Python package. Pseudocode of the end-to-end pipeline:

```
01_build_features.py
    raw = load_uci('data/raw/data.txt')
    short_gaps, long_gaps = classify_missing(raw)
    cleaned = forward_fill(raw, max_gap=3 min)        # past-only
    hourly  = resample(cleaned, '1h', mean)
    features = build(target=hourly.Global_active_power,
                     lags=[1,24,168],
                     rolling=[3,24,168],
                     calendar=['hour','dow','month','is_weekend','is_french_holiday'],
                     cyclical=['hour','dow','month'])  # sin/cos
    save(features, 'data/features.parquet')

02_fetch_weather.py
    weather = meteostat.Hourly('07156', start, end).fetch()  # Paris-Montsouris
    save(weather[['temp','rhum','wspd']] + lag24(temp),
         'data/paris_montsouris_weather.parquet')

04..08  run_.py
    for fold in rolling_origin_splits(features.index, n=6, valid_days=30):
        Xt, yt = features.loc[fold.train], target.loc[fold.train]
        Xv, yv = features.loc[fold.valid], target.loc[fold.valid]
        model = Forecaster(**hyperparams)
        model.fit(Xt, yt)
        preds = model.predict_one_step_ahead(Xv)
        save(preds, f'results/_predictions_fold{fold.id}.csv')
        save(metrics(yv, preds), f'results/_fold_results.csv')

09_run_stacking.py
    base = load_predictions(['xgboost','sarima','lstm','gru'], folds=1..5)
    meta = Ridge(alpha=1.0).fit(base.X, base.y)
    eval(meta, predictions(folds=6))

10_run_dm_test_and_segments.py
    p_values = pairwise_dm_test(predictions_folds_1_6)
    segment_mae = group_by(['hour','dow','month'], MAE)
    save(p_values, 'results/dm_test_pvalues.csv')

11_aggregate_leaderboard.py
    combine fold_results CSVs -> mean +- std per model -> results/leaderboard.csv

12_build_technical_report.py / 13_build_presentation.py
    load tracked artifacts, render PDF / PPTX
```

Appendix B — Author Contribution and Generative AI Statement

B.1 Author contribution (CRediT) — *DRAFT*

Draft. Using the CRediT taxonomy, contributions across the four authors are as follows:

- **Linda Carla Zorzoli & Mariavittoria Giurato** — *Investigation* (exploratory data analysis), *Resources* (research on documented French / European events explaining the dataset outages), *Validation* (review of the submission materials against the project rules).
- **Thierry Ishimwe & Cesar Dushimimana** — *Methodology* and *Software* (feature engineering, modelling, evaluation pipeline, reporting tooling); *Writing — original draft* and *Writing — review & editing*.
- **Conceptualization** and **Project administration** were shared across all four authors.

B.2 Generative AI Statement — *DRAFT*

*Draft. We used Anthropic's Claude during development for: **(a) extensive leakage checking** — auditing the feature pipeline for same-timestamp target components, verifying that lag and rolling features are strictly past-only, and surfacing the bilateral-interpolation leak in the missing-value policy which we then fixed; **(b)** brainstorming the registry-based leakage-prevention design and the past-only-fill regression test; **(c)** reviewing wording across the notebook, this technical report, and the presentation. All AI-generated code was reviewed, executed, and validated against tests. All design decisions and final wording are owned and understood by the authors.*